In [27]:
```
#First-Order Derivative
#Find the first derivative of
#f(x) = x+2
from sympy import *

x = symbols('x')
f = x + 2

f1 = diff(f, x)
print(" fx =", f1)

display(Eq(Symbol('fx'), f1))
```

```
 fx = 1
```

$$fx = 1$$

In [26]:
```
from sympy import *

x = symbols('x')
f = x**3 + 2*x**2 + 5*x + 7

f1 = diff(f, x)
print("fx =", f1)

display(Eq(Symbol('fx'), f1))
```

```
fx = 3*x**2 + 4*x + 5
```

$$fx = 3x^2 + 4x + 5$$

In [25]:
```
#Second and Third Order Derivatives
from sympy import *

x = symbols('x')
f = sin(x) + x**2

f2 = diff(f, x, 2)
f3 = diff(f, x, 3)

print("Second Derivative fxx =", f2)
print("Third Derivative fxxx =", f3)

display(Eq(Symbol('fx'), f2))
display(Eq(Symbol('fy'), f3))
```

```
Second Derivative fxx = 2 - sin(x)
Third Derivative fxxx = -cos(x)
```

$$fx = 2 - \sin(x)$$

$$fy = -\cos(x)$$

In [24]:
```
#Partial Derivatives
from sympy import *

x, y = symbols('x y')
f = x**2 * y + 3*x*y**2

f_x = diff(f, x)
f_y = diff(f, y)

print("fx =", fx)
print("fy =", fy)

display(Eq(Symbol('fx'), f_x))
display(Eq(Symbol('fy'), f_y))
```

```
fx = 2*x*y + 3*y**2
fy = x**2 + 6*x*y
```

$$fx = 2xy + 3y^2$$

$$fy = x^2 + 6xy$$

In [23]:
```
#Second-Order Partial Derivatives
from sympy import *

x, y = symbols('x y')
f = x**3*y**2 + 4*x*y

f_xx = diff(f, x, 2)
f_yy = diff(f, y, 2)
f_xy = diff(f, x, y)

print("fxx =", f_xx)
print("fyy =", f_yy)
print("fxy =", f_xy)

display(Eq(Symbol('fxx'), f_xx))
display(Eq(Symbol('fyy'), f_yy))
display(Eq(Symbol('fxy'), f_xy))
```

```
fxx = 6*x*y**2
fyy = 2*x**3
fxy = 2*(3*x**2*y + 2)
```

$$fxx = 6xy^2$$

$$fyy = 2x^3$$

$$fxy = 2 \cdot \left(3x^2y + 2\right)$$

In [22]:

```python
#Second-Order Partial Derivatives
from sympy import *

x, y, z = symbols('x y z')
f = x**3*y**2 + 4*x*y + 2*z

f_xx = diff(f, x, 2)
f_yy = diff(f, y, 2)
f_zz = diff(f, z, 2)
f_xyz = diff(f, x, y, z)

print("fxx =", f_xx)
print("fyy =", f_yy)
print("fzz =", f_zz)
print("fxyz =", f_xyz)
display(Eq(Symbol('fxx'), f_xx))
display(Eq(Symbol('fyy'), f_yy))
display(Eq(Symbol('fzz'), f_zz))
display(Eq(Symbol('fxyz'), f_xyz))
```

```
fxx = 6*x*y**2
fyy = 2*x**3
fzz = 0
fxyz = 0
```

$$fxx = 6xy^2$$

$$fyy = 2x^3$$

$$fzz = 0$$

$$fxyz = 0$$

In [30]:
```
#Third-Order Partial Derivatives
from sympy import *

x, y, z = symbols('x y z')
f = x**3*y**2 + 4*x*y + 2*z

f_xxx = diff(f, x, 3)
f_yyy = diff(f, y, 3)
f_zzz = diff(f, z, 3)
f_xyz = diff(f, x, y, z)

print("fxxx =", f_xxx)
print("fyyy =", f_yyy)
print("fzzz =", f_zzz)
print("fxyz =", f_xyz)

display(Eq(Symbol('fxxx'), f_xxx))
display(Eq(Symbol('fyyy'), f_yyy))
display(Eq(Symbol('fzzz'), f_zzz))
display(Eq(Symbol('fxyz'), f_xyz))
```

```
fxxx = 6*y**2
fyyy = 0
fzzz = 0
fxyz = 0
```

$$fxxx = 6y^2$$

$$fyyy = 0$$

$$fzzz = 0$$

$$fxyz = 0$$

In [21]:
```
#Third-Order Partial Derivatives
from sympy import *

x, y, z = symbols('x y z')
f = sin(x*y*z)

f_xxx = diff(f, x, 3)
f_yyy = diff(f, y, 3)
f_zzz = diff(f, z, 3)
f_xyz = diff(f, x, y, z)

print("fxxx =", f_xxx)
print("fyyy =", f_yyy)
print("fzzz =", f_zzz)
print("fxyz =", f_xyz)

display(Eq(Symbol('fxxx'), f_xxx))
display(Eq(Symbol('fyyy'), f_yyy))
display(Eq(Symbol('fzzz'), f_zzz))
display(Eq(Symbol('fxyz'), f_xyz))
```

```
fxxx = -y**3*z**3*cos(x*y*z)
fyyy = -x**3*z**3*cos(x*y*z)
fzzz = -x**3*y**3*cos(x*y*z)
fxyz = -x**2*y**2*z**2*cos(x*y*z) - 3*x*y*z*sin(x*y*z) + cos(x*y*z)
```

$$fxxx = -y^3 z^3 \cos(xyz)$$

$$fyyy = -x^3 z^3 \cos(xyz)$$

$$fyyy = -x^3 z^3 \cos(xyz)$$

$$fxyz = -x^2 y^2 z^2 \cos(xyz) - 3xyz \sin(xyz) + \cos(xyz)$$

In [28]:
```
#Third-Order Partial Derivatives
from sympy import *

x, y, z = symbols('x y z')
f = x + y + z

f_xxx = diff(f, x, 3)
f_yyy = diff(f, y, 3)
f_zzz = diff(f, z, 3)
f_xyz = diff(f, x, y, z)

print("fxxx =", f_xxx)
print("fyyy =", f_yyy)
print("fzzz =", f_zzz)
print("fxyz =", f_xyz)

display(Eq(Symbol('fxxx'), f_xxx))
display(Eq(Symbol('fyyy'), f_yyy))
display(Eq(Symbol('fzzz'), f_zzz))
display(Eq(Symbol('fxyz'), f_xyz))
```

```
fxxx = 0
fyyy = 0
fzzz = 0
fxyz = 0
```

$$fxxx = 0$$

$$fyyy = 0$$

$$fyyy = 0$$

$$fxyz = 0$$

In [31]:
```
#Third-Order Partial Derivatives
from sympy import *

x, y, z = symbols('x y z')
f = x**2 + y**2 + z**2

f_x = diff(f, x, 1)
f_y = diff(f, y, 1)
f_z = diff(f, z, 1)

f_xx = diff(f, x, 2)
f_yy = diff(f, y, 2)
f_zz = diff(f, z, 2)

f_xxx = diff(f, x, 3)
f_yyy = diff(f, y, 3)
f_zzz = diff(f, z, 3)
f_xyz = diff(f, x, y, z)

display(Eq(Symbol('fx'), f_x))
display(Eq(Symbol('fy'), f_y))
display(Eq(Symbol('fz'), f_z))


display(Eq(Symbol('fxx'), f_xx))
display(Eq(Symbol('fyy'), f_yy))
display(Eq(Symbol('fzz'), f_zz))
display(Eq(Symbol('fxyz'), f_xyz))


display(Eq(Symbol('fxxx'), f_xxx))
display(Eq(Symbol('fyyy'), f_yyy))
display(Eq(Symbol('fzzz'), f_zzz))
display(Eq(Symbol('fxyz'), f_xyz))
```

$$fx = 2x$$

$$fy = 2y$$

$$fz = 2z$$

$$fxx = 2$$

$$fyy = 2$$

$$fzz = 2$$

$$fxyz = 0$$

$$fxxx = 0$$

$$fyyy = 0$$

$$fzzz = 0$$

$$fxyz = 0$$

In [32]:
```
#Third-Order Partial Derivatives
from sympy import *

x, y, z = symbols('x y z')
f = x**3 + y**3 + z**3

f_x = diff(f, x, 1)
f_y = diff(f, y, 1)
f_z = diff(f, z, 1)

f_xx = diff(f, x, 2)
f_yy = diff(f, y, 2)
f_zz = diff(f, z, 2)

f_xxx = diff(f, x, 3)
f_yyy = diff(f, y, 3)
f_zzz = diff(f, z, 3)
f_xyz = diff(f, x, y, z)

display(Eq(Symbol('fx'), f_x))
display(Eq(Symbol('fy'), f_y))
display(Eq(Symbol('fz'), f_z))


display(Eq(Symbol('fxx'), f_xx))
display(Eq(Symbol('fyy'), f_yy))
display(Eq(Symbol('fzz'), f_zz))
display(Eq(Symbol('fxyz'), f_xyz))


display(Eq(Symbol('fxxx'), f_xxx))
display(Eq(Symbol('fyyy'), f_yyy))
display(Eq(Symbol('fzzz'), f_zzz))
display(Eq(Symbol('fxyz'), f_xyz))
```

$$fx = 3x^2$$

$$fy = 3y^2$$

$$fz = 3z^2$$

$$fxx = 6x$$

$$fyy = 6y$$

$$fzz = 6z$$

$$fxyz = 0$$

$$fxxx = 6$$

$$fyyy = 6$$

$$fzzz = 6$$

$$fxyz = 0$$

In [33]:
```
#Third-Order Partial Derivatives
from sympy import *

x, y, z = symbols('x y z')
f = x**2*y + y*z**3

f_x = diff(f, x, 1)
f_y = diff(f, y, 1)
f_z = diff(f, z, 1)

f_xx = diff(f, x, 2)
f_yy = diff(f, y, 2)
f_zz = diff(f, z, 2)

f_xxx = diff(f, x, 3)
f_yyy = diff(f, y, 3)
f_zzz = diff(f, z, 3)
f_xyz = diff(f, x, y, z)

display(Eq(Symbol('fx'), f_x))
display(Eq(Symbol('fy'), f_y))
display(Eq(Symbol('fz'), f_z))


display(Eq(Symbol('fxx'), f_xx))
display(Eq(Symbol('fyy'), f_yy))
display(Eq(Symbol('fzz'), f_zz))
display(Eq(Symbol('fxyz'), f_xyz))


display(Eq(Symbol('fxxx'), f_xxx))
display(Eq(Symbol('fyyy'), f_yyy))
display(Eq(Symbol('fzzz'), f_zzz))
display(Eq(Symbol('fxyz'), f_xyz))
```

$$fx = 2xy$$

$$fy = x^2 + z^3$$

$$fz = 3yz^2$$

$$fxx = 2y$$

$$fyy = 0$$

$$fzz = 6yz$$

$$fxyz = 0$$

$$fxxx = 0$$

$$fyyy = 0$$

$$fzzz = 6y$$

$$fxyz = 0$$

In [35]:

```
#Third-Order Partial Derivatives
from sympy import *

x, y, z = symbols('x y z')
f = x*sin(2*y + 3*z)

f_x = diff(f, x, 1)
f_y = diff(f, y, 1)
f_z = diff(f, z, 1)

f_xx = diff(f, x, 2)
f_yy = diff(f, y, 2)
f_zz = diff(f, z, 2)

f_xxx = diff(f, x, 3)
f_yyy = diff(f, y, 3)
f_zzz = diff(f, z, 3)
f_xyz = diff(f, x, y, z)

display(Eq(Symbol('fx'), f_x))
display(Eq(Symbol('fy'), f_y))
display(Eq(Symbol('fz'), f_z))


display(Eq(Symbol('fxx'), f_xx))
display(Eq(Symbol('fyy'), f_yy))
display(Eq(Symbol('fzz'), f_zz))
display(Eq(Symbol('fxyz'), f_xyz))


display(Eq(Symbol('fxxx'), f_xxx))
display(Eq(Symbol('fyyy'), f_yyy))
display(Eq(Symbol('fzzz'), f_zzz))
display(Eq(Symbol('fxyz'), f_xyz))
```

$$fx = \sin(2y + 3z)$$

$$fy = 2x\cos(2y + 3z)$$

$$fz = 3x\cos(2y + 3z)$$

$$fxx = 0$$

$$fyy = -4x\sin(2y + 3z)$$

$$fzz = -9x\sin(2y + 3z)$$

$$fxyz = -6\sin(2y + 3z)$$

$$fxxx = 0$$

$$fyyy = -8x\cos(2y + 3z)$$

$$fzzz = -27x\cos(2y + 3z)$$

$$fxyz = -6\sin(2y + 3z)$$

In [36]:
```
#Third-Order Partial Derivatives
from sympy import *

x, y, z = symbols('x y z')
f = sin(x**3*y**3*z**3)

f_x = diff(f, x, 1)
f_y = diff(f, y, 1)
f_z = diff(f, z, 1)

f_xx = diff(f, x, 2)
f_yy = diff(f, y, 2)
f_zz = diff(f, z, 2)

f_xxx = diff(f, x, 3)
f_yyy = diff(f, y, 3)
f_zzz = diff(f, z, 3)
f_xyz = diff(f, x, y, z)

display(Eq(Symbol('fx'), f_x))
display(Eq(Symbol('fy'), f_y))
display(Eq(Symbol('fz'), f_z))


display(Eq(Symbol('fxx'), f_xx))
display(Eq(Symbol('fyy'), f_yy))
display(Eq(Symbol('fzz'), f_zz))
display(Eq(Symbol('fxyz'), f_xyz))


display(Eq(Symbol('fxxx'), f_xxx))
display(Eq(Symbol('fyyy'), f_yyy))
display(Eq(Symbol('fzzz'), f_zzz))
display(Eq(Symbol('fxyz'), f_xyz))
```

$$fx = 3x^2y^3z^3\cos\left(x^3y^3z^3\right)$$

$$fy = 3x^3y^2z^3\cos\left(x^3y^3z^3\right)$$

$$fz = 3x^3y^3z^2\cos\left(x^3y^3z^3\right)$$

$$fxx = 3xy^3z^3\left(-3x^3y^3z^3\sin\left(x^3y^3z^3\right) + 2\cos\left(x^3y^3z^3\right)\right)$$

$$fyy = 3x^3yz^3\left(-3x^3y^3z^3\sin\left(x^3y^3z^3\right) + 2\cos\left(x^3y^3z^3\right)\right)$$

$$fzz = 3x^3y^3z\left(-3x^3y^3z^3\sin\left(x^3y^3z^3\right) + 2\cos\left(x^3y^3z^3\right)\right)$$

$$fxyz = 27x^2y^2z^2\left(-x^6y^6z^6\cos\left(x^3y^3z^3\right) - 3x^3y^3z^3\sin\left(x^3y^3z^3\right) + \cos\left(x^3y^3z^3\right)\right)$$

$$fxxx = 3y^3z^3\left(-9x^6y^6z^6\cos\left(x^3y^3z^3\right) - 18x^3y^3z^3\sin\left(x^3y^3z^3\right) + 2\cos\left(x^3y^3z^3\right)\right)$$

$$fyyy = 3x^3z^3\left(-9x^6y^6z^6\cos\left(x^3y^3z^3\right) - 18x^3y^3z^3\sin\left(x^3y^3z^3\right) + 2\cos\left(x^3y^3z^3\right)\right)$$

$$fzzz = 3x^3y^3\left(-9x^6y^6z^6\cos\left(x^3y^3z^3\right) - 18x^3y^3z^3\sin\left(x^3y^3z^3\right) + 2\cos\left(x^3y^3z^3\right)\right)$$

$$fxyz = 27x^2y^2z^2\left(-x^6y^6z^6\cos\left(x^3y^3z^3\right) - 3x^3y^3z^3\sin\left(x^3y^3z^3\right) + \cos\left(x^3y^3z^3\right)\right)$$

In [37]:
```python
#Third-Order Partial Derivatives
from sympy import *

x, y, z = symbols('x y z')
f = x*sin(2*y + 3*z)/cos(x*y*z)

f_x = diff(f, x, 1)
f_y = diff(f, y, 1)
f_z = diff(f, z, 1)

f_xx = diff(f, x, 2)
f_yy = diff(f, y, 2)
f_zz = diff(f, z, 2)

f_xxx = diff(f, x, 3)
f_yyy = diff(f, y, 3)
f_zzz = diff(f, z, 3)
f_xyz = diff(f, x, y, z)

display(Eq(Symbol('fx'), f_x))
display(Eq(Symbol('fy'), f_y))
display(Eq(Symbol('fz'), f_z))


display(Eq(Symbol('fxx'), f_xx))
display(Eq(Symbol('fyy'), f_yy))
display(Eq(Symbol('fzz'), f_zz))
display(Eq(Symbol('fxyz'), f_xyz))


display(Eq(Symbol('fxxx'), f_xxx))
display(Eq(Symbol('fyyy'), f_yyy))
display(Eq(Symbol('fzzz'), f_zzz))
display(Eq(Symbol('fxyz'), f_xyz))
```

$$fx = \frac{xyz \sin(xyz) \sin(2y+3z)}{\cos^2(xyz)} + \frac{\sin(2y+3z)}{\cos(xyz)}$$

$$fy = \frac{x^2 z \sin(xyz) \sin(2y+3z)}{\cos^2(xyz)} + \frac{2x \cos(2y+3z)}{\cos(xyz)}$$

$$fz = \frac{x^2 y \sin(xyz) \sin(2y+3z)}{\cos^2(xyz)} + \frac{3x \cos(2y+3z)}{\cos(xyz)}$$

$$fxx = \frac{yz\left(xyz\left(\frac{2\sin^2(xyz)}{\cos^2(xyz)} + 1\right) + \frac{2\sin(xyz)}{\cos(xyz)}\right)\sin(2y+3z)}{\cos(xyz)}$$

$$fyy = \frac{x\left(x^2z^2 \cdot \left(\frac{2\sin^2(xyz)}{\cos^2(xyz)} + 1\right)\sin(2y+3z) + \frac{4xz\sin(xyz)\cos(2y+3z)}{\cos(xyz)} - 4\sin(2y+3z)\right)}{\cos(xyz)}$$

$$fzz = \frac{x\left(x^{2}y^{2}\cdot\left(\frac{2\sin^{2}(xyz)}{\cos^{2}(xyz)}+1\right)\sin(2y+3z)+\frac{6xy\sin(xyz)\cos(2y+3z)}{\cos(xyz)}-9\sin(2y+3z)\right)}{\cos(xyz)}$$

$$fxyz = \frac{\begin{array}{l}\frac{6x^{3}y^{2}z^{2}\sin^{3}(xyz)\sin(2y+3z)}{\cos^{3}(xyz)}+\frac{5x^{3}y^{2}z^{2}\sin(xyz)\sin(2y+3z)}{\cos(xyz)}+\frac{4x^{2}y^{2}z\sin^{2}(xyz)\cos(2y+3z)}{\cos^{2}(xyz)}+2x^{2}\\(2y+3z)+\frac{8x^{2}yz\sin^{2}(xyz)\sin(2y+3z)}{\cos^{2}(xyz)}+4x^{2}yz\sin(2y+3z)-\frac{6xyz\sin(xyz)\sin(2y+3z)}{\cos(xyz)}\\+\frac{2x\sin(xyz)\sin(2y+3z)}{\cos(xyz)}-6\sin(2y+3z)\end{array}}{\cos(xyz)}$$

$$fxxx = \frac{y^{2}z^{2}\left(\frac{xyz\left(\frac{6\sin^{2}(xyz)}{\cos^{2}(xyz)}+5\right)\sin(xyz)}{\cos(xyz)}+\frac{6\sin^{2}(xyz)}{\cos^{2}(xyz)}+3\right)\sin(2y+3z)}{\cos(xyz)}$$

$$fyyy = \frac{x\left(\frac{x^{3}z^{3}\cdot\left(\frac{6\sin^{2}(xyz)}{\cos^{2}(xyz)}+5\right)\sin(xyz)\sin(2y+3z)}{\cos(xyz)}+6x^{2}z^{2}\cdot\left(\frac{2\sin^{2}(xyz)}{\cos^{2}(xyz)}+1\right)\cos(2y+3z)-\frac{1}{}\right.}{\cos(xyz)}$$

$$fzzz = \frac{x\left(\frac{x^{3}y^{3}\cdot\left(\frac{6\sin^{2}(xyz)}{\cos^{2}(xyz)}+5\right)\sin(xyz)\sin(2y+3z)}{\cos(xyz)}+9x^{2}y^{2}\cdot\left(\frac{2\sin^{2}(xyz)}{\cos^{2}(xyz)}+1\right)\cos(2y+3z)-\frac{2}{}\right.}{\cos(xyz)}$$

$$fxyz = \frac{\begin{array}{l}\frac{6x^{3}y^{2}z^{2}\sin^{3}(xyz)\sin(2y+3z)}{\cos^{3}(xyz)}+\frac{5x^{3}y^{2}z^{2}\sin(xyz)\sin(2y+3z)}{\cos(xyz)}+\frac{4x^{2}y^{2}z\sin^{2}(xyz)\cos(2y+3z)}{\cos^{2}(xyz)}+2x^{2}\\(2y+3z)+\frac{8x^{2}yz\sin^{2}(xyz)\sin(2y+3z)}{\cos^{2}(xyz)}+4x^{2}yz\sin(2y+3z)-\frac{6xyz\sin(xyz)\sin(2y+3z)}{\cos(xyz)}\\+\frac{2x\sin(xyz)\sin(2y+3z)}{\cos(xyz)}-6\sin(2y+3z)\end{array}}{\cos(xyz)}$$

In [38]:
```
#Third-Order Partial Derivatives
from sympy import *

x, y, z = symbols('x y z')
f = tan(x*y*z)

f_x = diff(f, x, 1)
f_y = diff(f, y, 1)
f_z = diff(f, z, 1)

f_xx = diff(f, x, 2)
f_yy = diff(f, y, 2)
f_zz = diff(f, z, 2)

f_xxx = diff(f, x, 3)
f_yyy = diff(f, y, 3)
f_zzz = diff(f, z, 3)
f_xyz = diff(f, x, y, z)

display(Eq(Symbol('fx'), f_x))
display(Eq(Symbol('fy'), f_y))
display(Eq(Symbol('fz'), f_z))


display(Eq(Symbol('fxx'), f_xx))
display(Eq(Symbol('fyy'), f_yy))
display(Eq(Symbol('fzz'), f_zz))
display(Eq(Symbol('fxyz'), f_xyz))


display(Eq(Symbol('fxxx'), f_xxx))
display(Eq(Symbol('fyyy'), f_yyy))
display(Eq(Symbol('fzzz'), f_zzz))
display(Eq(Symbol('fxyz'), f_xyz))
```

$$fx = yz\left(\tan^2(xyz) + 1\right)$$

$$fy = xz\left(\tan^2(xyz) + 1\right)$$

$$fz = xy\left(\tan^2(xyz) + 1\right)$$

$$fxx = 2y^2z^2\left(\tan^2(xyz) + 1\right)\tan(xyz)$$

$$fyy = 2x^2z^2\left(\tan^2(xyz) + 1\right)\tan(xyz)$$

$$fzz = 2x^2y^2\left(\tan^2(xyz) + 1\right)\tan(xyz)$$

$$fxyz = 2x^2y^2z^2\left(\tan^2(xyz) + 1\right)^2 + 4x^2y^2z^2\left(\tan^2(xyz) + 1\right)\tan^2(xyz) + 6xyz\left(\tan^2(x\right.$$

$$fxxx = 2y^3z^3\left(\tan^2(xyz) + 1\right)\left(3\tan^2(xyz) + 1\right)$$

$$fyyy = 2x^3z^3\left(\tan^2(xyz) + 1\right)\left(3\tan^2(xyz) + 1\right)$$

$$fzzz = 2x^3y^3\left(\tan^2(xyz) + 1\right)\left(3\tan^2(xyz) + 1\right)$$

$$fxyz = 2x^2y^2z^2\left(\tan^2(xyz) + 1\right)^2 + 4x^2y^2z^2\left(\tan^2(xyz) + 1\right)\tan^2(xyz) + 6xyz\left(\tan^2(x\right.$$

In [43]:

```
#Chain Rule

from sympy import *

x, y, t = symbols('x y t')
f = x**2 * y + y**2

# Substitute x(t) and y(t)
x_t = t**2
y_t = 3*t

# Partial derivatives
df_dx = diff(f, x)
df_dy = diff(f, y)

# Total derivative using chain rule
df_dt = df_dx * diff(x_t, t) + df_dy * diff(y_t, t)

# Substitute x(t), y(t) into expression
df_dt = simplify(df_dt.subs({x: x_t, y: y_t}))

display(Eq(Symbol('df/dt'), df_dt))
```

$$df/dt = 15t^4 + 18t$$

In [45]:

```
#Chain Rule

from sympy import *

x, y, t = symbols('x y t')
f = x**3 + y**3

# Substitute x(t) and y(t)
x_t = t**2
y_t = 3*t

# Partial derivatives
df_dx = diff(f, x)
df_dy = diff(f, y)

# Total derivative using chain rule
df_dt = df_dx * diff(x_t, t) + df_dy * diff(y_t, t)

# Substitute x(t), y(t) into expression
df_dt = simplify(df_dt.subs({x: x_t, y: y_t}))

display(Eq(Symbol('df/dt'), df_dt))
```

$$df/dt = t^2 \cdot \left(6t^3 + 81\right)$$

In [44]:
```
#Chain Rule

from sympy import *

x, y, t = symbols('x y t')
f = x**2 + y**2

# Substitute x(t) and y(t)
x_t = t**2
y_t = 3*t

# Partial derivatives
df_dx = diff(f, x)
df_dy = diff(f, y)

# Total derivative using chain rule
df_dt = df_dx * diff(x_t, t) + df_dy * diff(y_t, t)

# Substitute x(t), y(t) into expression
df_dt = simplify(df_dt.subs({x: x_t, y: y_t}))

display(Eq(Symbol('df/dt'), df_dt))
```

$$df/dt = 4t^3 + 18t$$

In [48]:
```
# 🤖 CardioBot (Case Study)
# This robot checks how blood flow changes with Pressure and Oxygen

from sympy import *

# Step 1: Make symbols for Pressure (P) and Oxygen (O)
P, O = symbols('P O')

# Step 2: Make a simple blood flow formula
# B = 2*P^2 + 3*O + 4
B = 2*P**2 + 3*O + 4

# Step 3: Find partial derivatives
dB_dP = diff(B, P)   # change with pressure
dB_dO = diff(B, O)   # change with oxygen

# Step 4: Give sample sensor readings
P_value = 6   # pressure reading
O_value = 7   # oxygen reading

# Step 5: Calculate blood flow and changes
flow = B.subs({P: P_value, O: O_value})
change_P = dB_dP.subs({P: P_value, O: O_value})
change_O = dB_dO.subs({P: P_value, O: O_value})

# Step 6: Show results
print("🤖 CardioBot Blood Flow Prediction")
print("----------------------------------")
print("Pressure =", P_value)
print("Oxygen =", O_value)
print("Blood Flow (B) =", flow)
print("Change due to Pressure (∂B/∂P) =", change_P)
print("Change due to Oxygen (∂B/∂O) =", change_O)

# Step 7: Give an easy message
if change_P > change_O:
    print("⚠️ Blood flow changes more with PRESSURE!")
elif change_O > change_P:
    print("⚠️ Blood flow changes more with OXYGEN!")
else:
    print("✅ Blood flow is balanced between Pressure and Oxygen!")
```

```
🤖 CardioBot Blood Flow Prediction
----------------------------------
Pressure = 6
Oxygen = 7
Blood Flow (B) = 97
Change due to Pressure (∂B/∂P) = 24
Change due to Oxygen (∂B/∂O) = 3
⚠️ Blood flow changes more with PRESSURE!
```